

Square ValuesAdditional Practice 7A.02.01 **ANSWER KEY**

1. Complete the table. Each area is the side length multiplied by itself.

Side s	6	7	8	9	10	11	12	13	14	15
Area s^2	36	49	64	81	100	121	144	169	196	225

225 is the largest perfect square the Grade 8 FAST will ask for, so this row is the ceiling. A student who writes 30 for $s = 15$ has multiplied by 2.

2. Select ALL of the numbers below that are perfect squares.

- | | |
|---|--|
| ☒ 4 | ☐ 12 |
| ☒ 25 | ☐ 32 |
| ☐ 50 | ☒ 81 |
| ☐ 90 | ☒ 121 |
| ☐ 150 | ☒ 169 |

Five of the ten. For each one selected, ask which whole number was squared: 2, 5, 9, 11, 13.

For questions 3 through 5, find the area of a square with the side length given.

3. Side length 15 units

$$\text{Area} = 15^2 = 15 \times 15 = 225 \text{ square units}$$

4. Side length $\frac{3}{4}$ meters

$$\text{Area} = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16} \text{ m}^2$$

Square the top and square the bottom. The answer is not 6/8.

5. Side length 1.2 units

$$\text{Area} = 1.2 \times 1.2 = 1.44 \text{ square units}$$

A student who answers 2.4 has doubled instead of squaring.

For questions 6 through 8, find the length of one side of a square with the area given.

6. Area 169 square inches

$13 \times 13 = 169$, so the side is 13 inches.

7. Area $\frac{9}{16}$ square meters

$3 \times 3 = 9$ and $4 \times 4 = 16$, so the side is $\frac{3}{4}$ m.

8. Area 0.64 square units

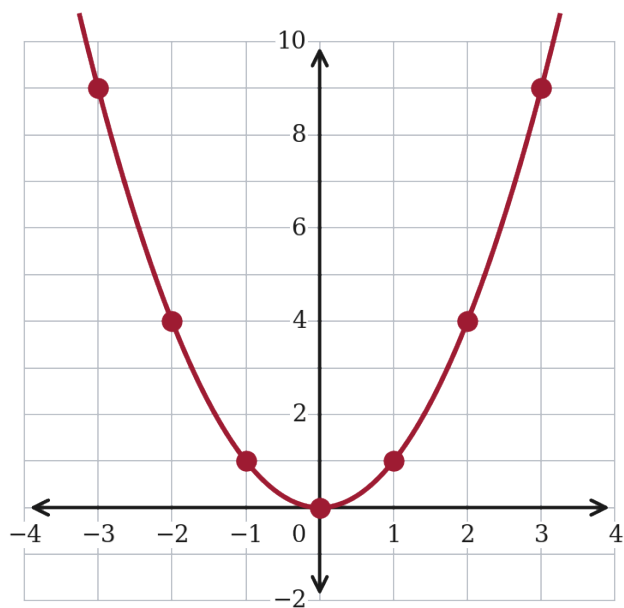
$0.8 \times 0.8 = 0.64$, so the side is 0.8 units.

A student who answers 0.08 has lost a decimal place. Squaring 0.08 gives 0.0064.

Questions 9 and 10 use the equation $y = x^2$.

9. Complete the table of values. Then plot the points on the grid.

x	Substitute into $y = x^2$	Coordinates (x, y)
-3	$y = (-3)^2 = 9$	$(-3, 9)$
-2	$y = (-2)^2 = 4$	$(-2, 4)$
-1	$y = (-1)^2 = 1$	$(-1, 1)$
0	$y = (0)^2 = 0$	$(0, 0)$
1	$y = (1)^2 = 1$	$(1, 1)$
2	$y = (2)^2 = 4$	$(2, 4)$
3	$y = (3)^2 = 9$	$(3, 9)$



The two ends, $(-3, 9)$ and $(3, 9)$, are the pair to point at: different x , same y .

10. A classmate says the graph of $y = x^2$ is always above the x-axis. Is the classmate right? Explain.

Not quite. The graph never goes below the x-axis, but at $x = 0$ it touches it.

$0 \times 0 = 0$, so the point $(0, 0)$ is on the x-axis, not above it.

This is the same distinction as question 10 of the worksheet, asked from the other direction. "Never below" is correct; "always above" is not.

11. Can a square have an area of 20 square units? If it can, name the two whole numbers its side length falls between. Explain how you know.

Yes. The side is between 4 and 5.

$4 \times 4 = 16$, which is too small, and $5 \times 5 = 25$, which is too big.

The side is not a whole number.

The point needs 4 and 5 named and both products given as the reason.